Research Brief: The Promise of AI Tutoring in Ghana: A Low-Cost Path to Learning Gains

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The Issue

Ghana faces a severe learning crisis. While enrollment has increased, many students are not acquiring foundational literacy and numeracy skills. This is compounded by teacher shortages, overcrowded classrooms, and a lack of individualized support, particularly in rural and underserved communities. Traditional EdTech solutions often require expensive hardware and reliable internet, making them inaccessible to the most vulnerable learners.

The Trend: AI Tutoring via Basic Mobile Phones

An emerging and highly promising trend is the deployment of AI-driven tutoring tools that operate on basic, low-tech mobile phones. Rather than requiring a laptop or high-speed internet, these tools use simple messaging platforms like WhatsApp or SMS to deliver personalized, interactive lessons and quizzes directly to students. This approach leverages the high penetration of basic mobile phones in Ghana to bypass infrastructure barriers.

The Evidence (Research-Backed)

Recent research and pilot programs demonstrate the immense potential of this approach:

- **The Rori Case Study (Ghana):** A pilot program using an AI tutor named "Rori" deployed via WhatsApp in Ghana showed that it produced learning gains equivalent to roughly an extra year of schooling at a marginal cost of about USD 5 per child (Africa Practice, 2026).
- **Interactive Satellite Instruction (Ghana):** A cluster randomized controlled trial by Johnston and Ksoll (2017) in Ghanaian primary schools found significant gains (p¡0.05) in rural students' numeracy and foundational literacy skills when live instruction was broadcast via satellite [citation:2][citation:9]. The study noted that these gains resulted from improved instructional quality rather than increased instruction time, suggesting that distance learning technologies can overcome geographical barriers when designed interactively.

- **The Limits of Low-Tech Interventions:** Not all technological interventions are successful. A randomized control trial in Sierra Leone found that live tutoring calls conducted during the COVID-19 pandemic did not improve learning outcomes [citation:3]. The study noted that the program had moderate fidelity but generated low engagement from parents and students, and the calls followed a set script rather than providing personalized instruction [citation:3]. This underscores that EdTech must be carefully designed to meet specific contextual needs and foster engagement.
- **The Usage Gap (Systematic Review):** A systematic review of 199 articles on Technology-Enhanced Learning (TEL) in African higher education found that while technology has increased access to education, its pedagogical application remains limited [citation:1][citation:8]. Using the SAMR framework, the authors found that most technology use in Africa is at the "substitution" and "augmentation" levels, while deep integration at the "modification" and "redefinition" levels is rare [citation:1]. This highlights a critical need for continuous professional development for educators and better-designed AI tools that fundamentally transform the learning process.

What This Means for Ghana (and the Company)

This trend has profound implications for the future of education in Ghana:

1. **Scalability and Cost-Effectiveness:** Because it relies on basic phones rather than expensive hardware, AI tutoring via SMS/WhatsApp can scale rapidly to reach the most marginalized students at a fraction of the cost of building new schools or hiring thousands of new teachers.
2. **Policy Implementation:** To fully realize this potential, Ghana needs an education-centered AI policy framework. This includes ensuring cultural and linguistic relevance (e.g., adapting AI tools to understand local dialects like Twi), protecting student data, and training teachers to supplement AI tutoring rather than being replaced by it.
3. **Addressing the Engagement Gap:** The failure of the Sierra Leone tutoring calls shows that simply delivering content is not enough [citation:3]. AI tutors must be interactive, personalized, and designed to keep students engaged over time, not just broadcast information.

Conclusion

AI tutoring over basic mobile infrastructure represents a paradigm shift. It moves EdTech from a luxury good for the privileged to a scalable, low-cost tool for achieving universal basic education in Ghana. However, to avoid the pitfalls of past interventions, these tools must be designed with pedagogical depth and local context in mind.

References

- Africa Practice. (2026). *AI Doesn't Need Perfect Classrooms: The Rori Case Study.*
- Beuermann, D. W., et al. (2023). Live tutoring calls did not improve learning during the COVID-19 pandemic in Sierra Leone. *Journal of Development Economics.* ScienceDirect
- Habarurema, J. B., Limone, P., Bikorimana, E., & Di Fuccio, R. (2024). Technology-Enhanced Learning in Africa: A Challenge and Opportunity? *Africa Education Review*, 20(6), 20-57. https://doi.org/10.1080/18146627.2024.2433982
- Johnston, J., & Ksoll, C. (2017). Effectiveness of Interactive Satellite-Transmitted Instruction: Experimental Evidence from Ghanaian Primary Schools. *Stanford Center for Education Policy Analysis (CEPA) Working Paper No. 17-08.* ERIC